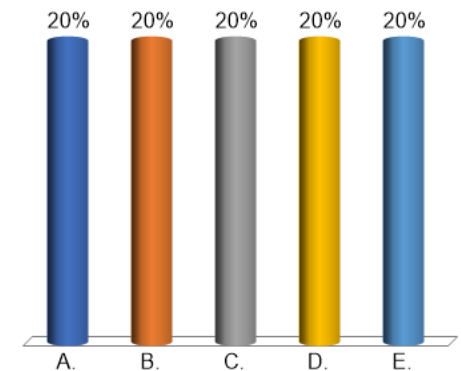


BPM1
CPR IMCQ 9
Physiology Key

A 68-year-old man comes to the physician for a follow-up examination. He has a 22-year history of chronic obstructive pulmonary disease. He has smoked 2 packs of cigarettes daily for 20 years. His blood pressure is 128/86 mmHg, and respirations are 20/min with prolonged expiration. Pulmonary examination shows hyperresonant percussion notes and decreased breath sounds in all lung fields. Cardiac examination shows parasternal movements and a loud S2 in the pulmonary area. An ECG shows right axis deviation. Pulmonary studies show an increased mean pulmonary arterial pressure. Which of the following mechanisms best explains the pulmonary study finding in this patient?

- A. Increased pulmonary capillary cross-sectional area
- B. Elevated pulmonary capillary hydrostatic pressure
- C. Hypoxic pulmonary vasoconstriction
- D. Decreased lung surfactants
- E. Elevated left atrial pressure



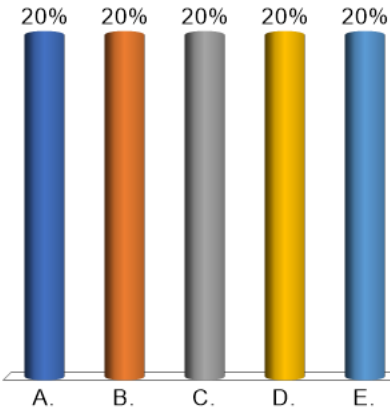
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SOM.MK.II.BPM1.3.CPR.2.PHYS.0337.Discuss pathophysiologic basis of clinical features, laboratory findings and complications in a patient with chronic obstructive lung disease

A 62-year-old woman is brought to the emergency department because of a 4-hour history of confusion and 3-day history of fever, cough and progressive shortness of breath. She has a 10-year history of chronic obstructive pulmonary disease. She was hospitalized twice with similar symptoms in the past year. Her temperature is 38.4C (101F), pulse is 86/min, blood pressure is 132/86 mmHg, and respirations are 24/min. Pulse oximetry on room air shows an oxygen saturation of 78% (Normal:>95%). Physical examination shows respiratory distress, cyanosis, and increased movements of chest and abdomen. Arterial blood gas analysis on room air shows a PaCO₂ of 80 mmHg (Normal:40-44) and a pH of 7.31. She is provided with supplemental oxygen through a face mask at a F_{IO₂} of 40%. Her condition improves. Which of the following sets of alveolar partial pressures of oxygen (PAO₂) is most likely in this patient before and after supplemental oxygen therapy? (Consider, Resp.Quotient, R = 0.8)

Option	Before Supplemental O2	After Supplemental O2
A	100 mm Hg	200 mm Hg
B	85 mm Hg	225 mm Hg
C	60 mm Hg	150 mm Hg
D	50 mm Hg	185 mm Hg
E	35 mm Hg	125 mm Hg



Use Equation: $PAO_2 = (760 - 47) \times FiO_2 - [PaCO_2 / R]$.

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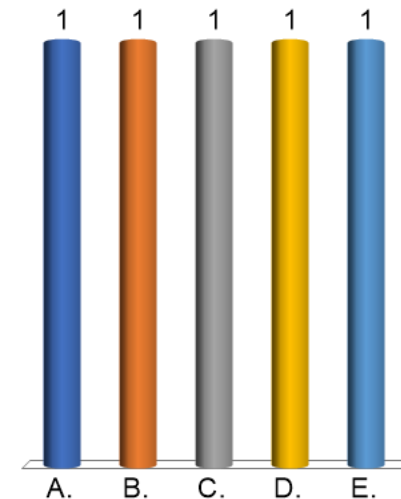
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SOM.MK.II.BPM1.3.CPR.2.PHY S.0238. Calculate the alveolar oxygen partial pressure (PAO₂) and the amount of supplemental O₂ required to overcome a reduction in PAO₂ caused by hypoventilation or high altitude using the simplified form of the alveolar gas equation.

Use Equation: $PAO_2 = (760 - 47) \times FiO_2 - [PaCO_2/R]$.

An 11-year-old girl is brought to the emergency department by her mother because of a 2-hour history of shortness of breath and fatigue, and 1-day history of nausea and vomiting. Two years ago, she was diagnosed with type 1 diabetes mellitus and is taking human insulin injections. The mother says she did not take insulin injection yesterday because of vomiting. Her temperature is 37° C (98.6° F), pulse is 122/min, and respirations are 24/min with increased tidal volume. Physical examination shows dry mucous membranes and altered mental status. Laboratory studies show elevated concentrations of glucose and ketone bodies in blood and urine. Arterial blood gas analysis on room air shows a PaCO₂ of 30 mm Hg (Normal: 40), pH of 7.32 (Normal: 7.4) and serum bicarbonate of 18 mEq/L (Normal: 24). She is diagnosed with diabetic ketoacidosis, a type of metabolic acidosis. Stimulation of which of the following is most likely responsible for an increased ventilatory response observed in this patient?

- A. Peripheral chemoreceptors by a low PaCO₂
- B. Peripheral chemoreceptors by a low blood pH
- C. Peripheral chemoreceptor by a low PaO₂
- D. Central chemoreceptors by a low blood pH
- E. Central chemoreceptor by a low PaO₂



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SOM.MK.II.BPM1.3.CPR.2.PHYS.0372. Differentiate the relative importance of each in sensing alterations in blood gases.

CPR IMCQ 9

HCB Questions

A 32-year-old man is brought to the emergency department because of a 30-minute history of generalized weakness after being involved in a motor vehicle collision. His blood pressure is 60/40 mm Hg and pulse is 115/min. Physical examination shows multiple traumatic injuries with pallor of skin and mucous membranes and weak peripheral pulses. Which of the following best characterizes the cells that will most likely be stimulated to release an enzyme for the eventual raising of the blood pressure in this patient?

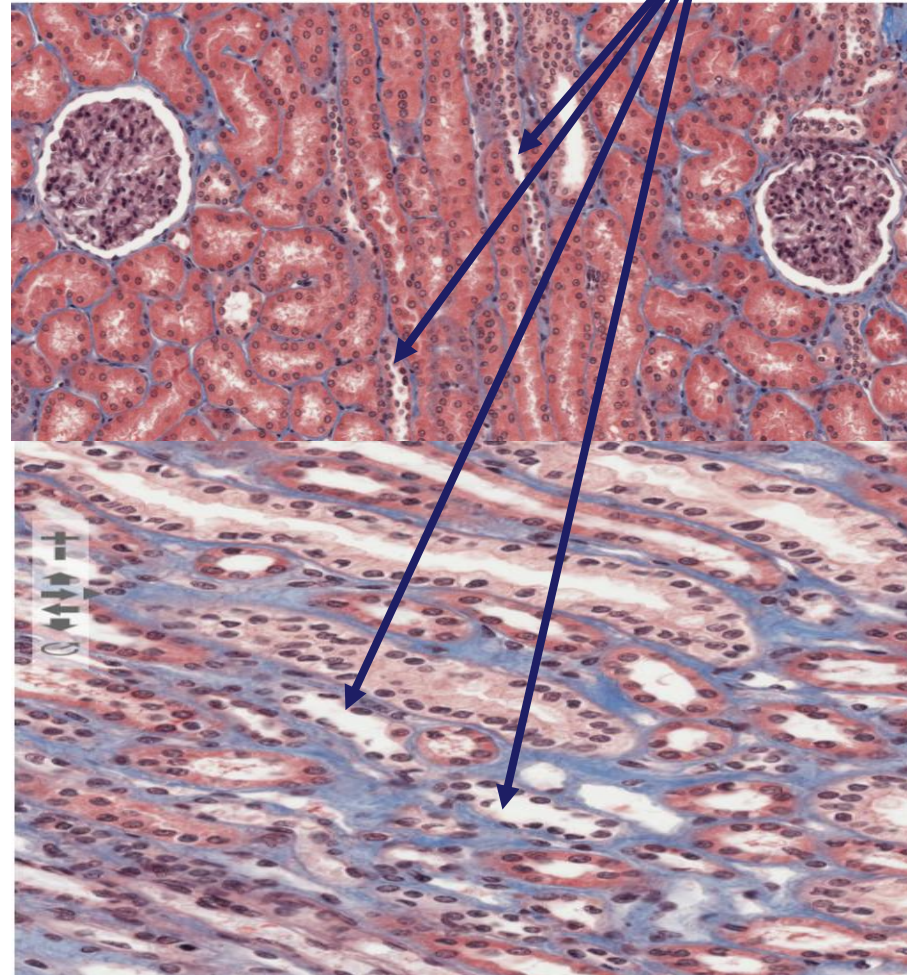
- A. Modified epithelial cells of the Bowman's capsule
- B. Epithelial cells of proximal convoluted tubule
- C. Endothelial cells of the glomerulus capillaries
- D. Mesangial cells within the glomerulus
- E. Modified tunica media cells of the afferent arteriole
- F. Principal cells of the collecting ducts
- G. Modified cells of the distal straight tubule

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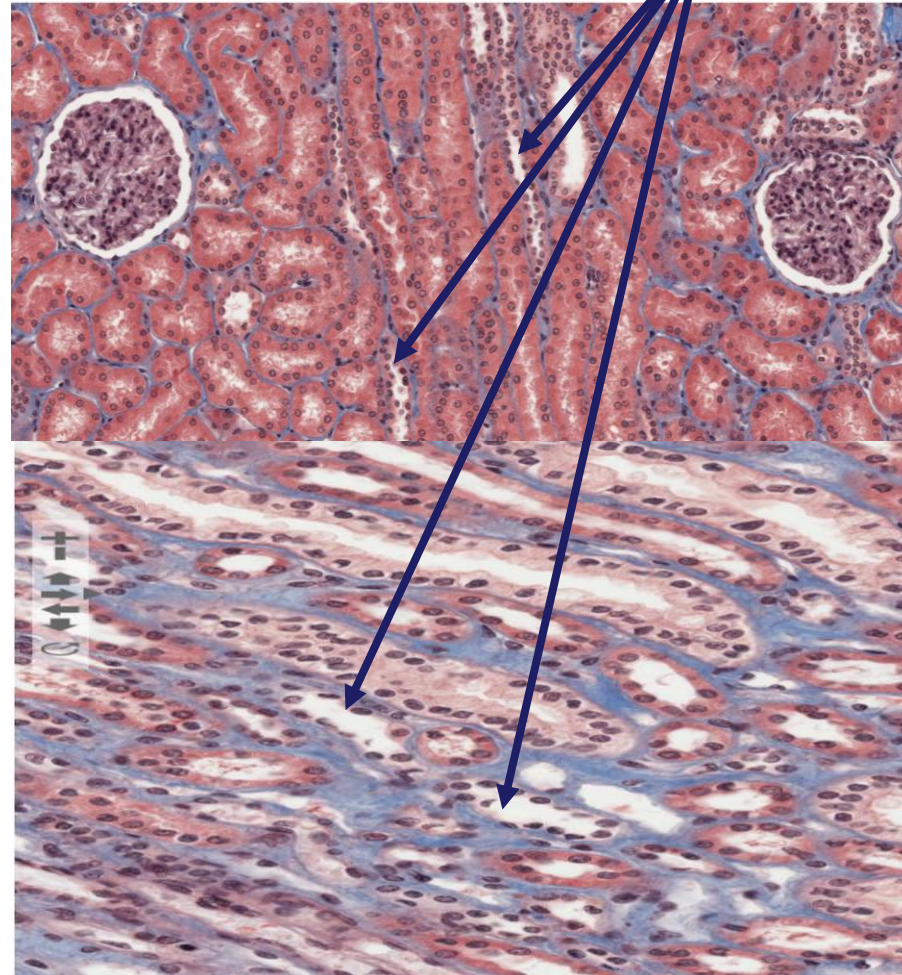
A 60-year-old man comes to the physician because of a 5-month history of pain and swelling of his right great toe. His vital signs are within normal limits. Physical examination shows warm, tender swelling of the right great toe with decreased active and passive range-of-motion. An x-ray of the right great toe shows lytic lesions within the proximal phalanges. Laboratory studies show elevated uric acid and decreased glomerular filtration rate. Renal imaging studies show atrophy of both kidneys. A renal biopsy is obtained and shows damage to the structures at the tips of the black arrows in the attached photomicrograph. Which of the following best characterizes the affected structure?

- A. Apically located epithelial nuclei
- B. Surrounded by the Bowman's capsule
- C. Well-developed brush border
- D. Cells have a primary cilium on its apical surface
- E. Apical part of the cells is covered by a urothelial plaque



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A 65-year-old woman comes to the physician because of a 3-month history of bright red blood in her urine and a 6-month history of an unintentional weight loss of 30 lbs (13.6 kgs). She has been menopausal for the past 10 years. A urinalysis shows **red** blood cells and leukocytes, indicative of a urinary tract infection. An ultrasound shows a mass in the right distal ureter **that** extends into the urinary bladder. A biopsy sample shows neoplastic changes to the epithelial cells that line these structure, resulting in loss of some of the functions of these cells. Which of the following best characterizes this lining epithelium of the affected structures?

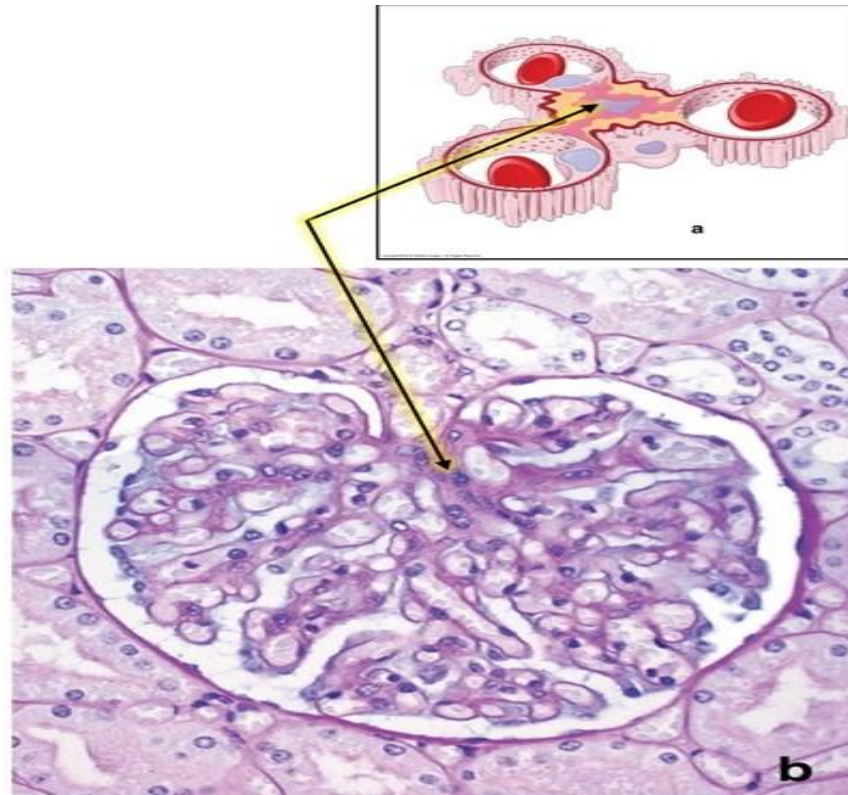
- A. Apical cells have junctional complexes that interact with intermediate filaments
- B. Apical cells have a crystalline array of uroplakin
- C. Apical cells can reabsorb water under the influence of antidiuretic hormone
- D. Apical cells have junctional complexes that facilitate communication between them
- E. Apical cells can reuptake calcium ions under the influence of parathyroid hormone

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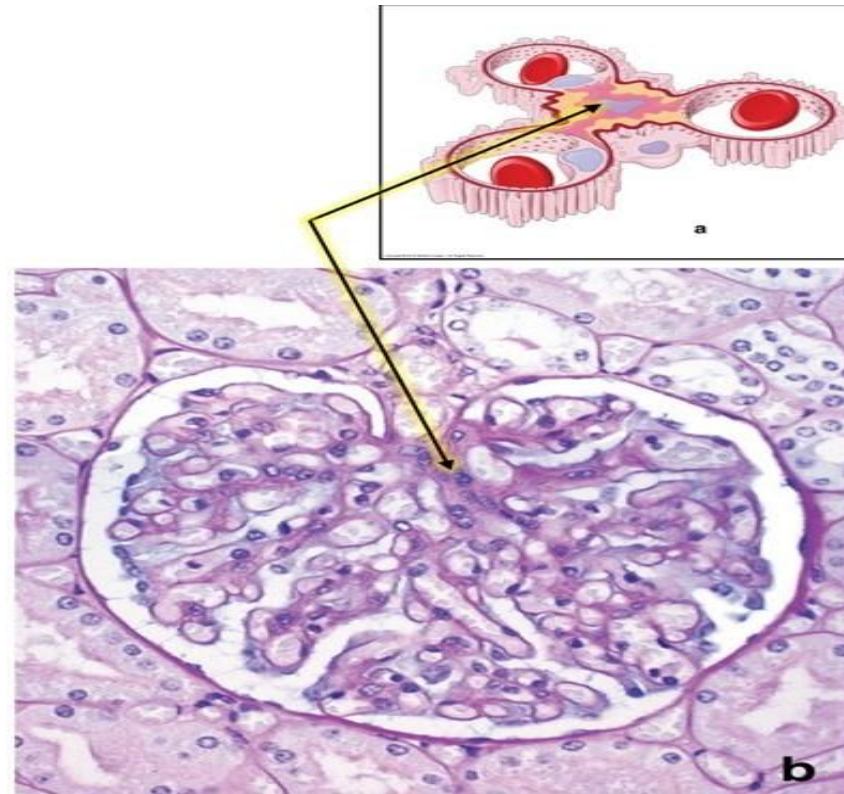
An 87-year-old woman dies from renal failure, despite all appropriate treatment. An autopsy specimen shows damage to the renal corpuscles. The main finding was proliferation of the cell shown in the attached micrograph. Which of the following is a normal function of these cells?

- A. Filtration of molecules smaller than 70,000 Da
- B. Phagocytosis of molecules trapped in filtration barrier
- C. Reabsorption of glucose
- D. Signaling from macula densa to juxtaglomerular cells
- E. Synthesis and secretion of atrial natriuretic factor



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A 42-year-old woman comes to the physician with a 1-week history of lower extremity edema and foamy urine. She has a history of hypertension and is currently taking medication for it. Physical examination shows bilateral pedal edema. Urinalysis shows proteinuria. Laboratory studies which include a renal function panel show elevated serum creatinine levels and a decreased glomerular filtration rate (GFR). A kidney biopsy is performed to evaluate the integrity of the glomerular filtration barrier. Which of the following is the most likely component of the glomerular filtration barrier responsible for preventing the passage of proteins from the blood into the urinary space?

- A. Fenestrations of the endothelial cells
- B. Mesangial cells with contractile properties
- C. Bowman's capsule lined by simple squamous epithelium
- D. Slit diaphragms between the pedicels of Podocytes
- E. Absence of heparan sulfate in the lamina rare externa

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A 34-year-old man comes to the physician with a 1-week history of voiding excessive urine and extreme thirst. Physical examination shows signs of mild dehydration. Laboratory tests show a serum electrolyte levels and urine osmolality levels that indicate impaired water reabsorption. Microscopic examination of a renal biopsy specimen shows abnormalities of the collecting ducts. Which of the following histological features best characterize these ducts in the region of the renal cortex?

- A. Well-developed brush border
- B. Simple squamous epithelium
- C. Absence of aquaporin-2 channels
- D. Presence of principal and intercalated cells
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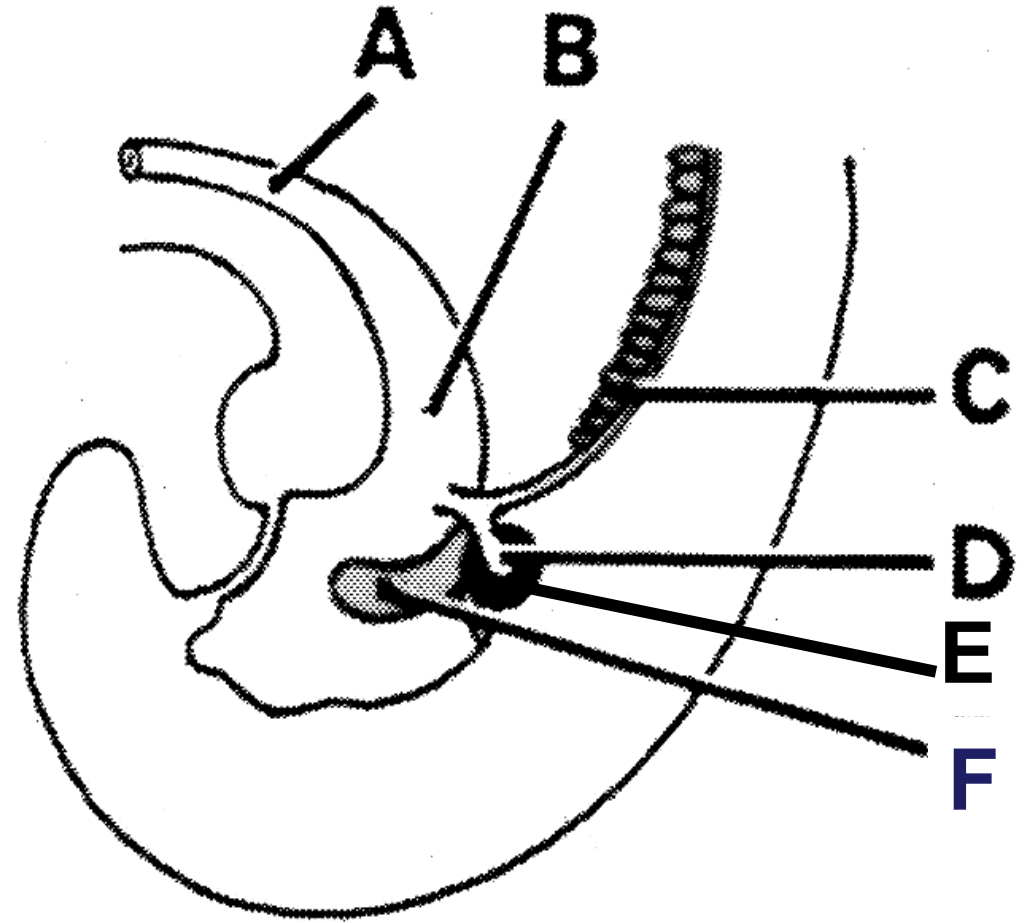
Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal (CPR) | IMCQ 9

CPR IMCQ 9 Anatomy

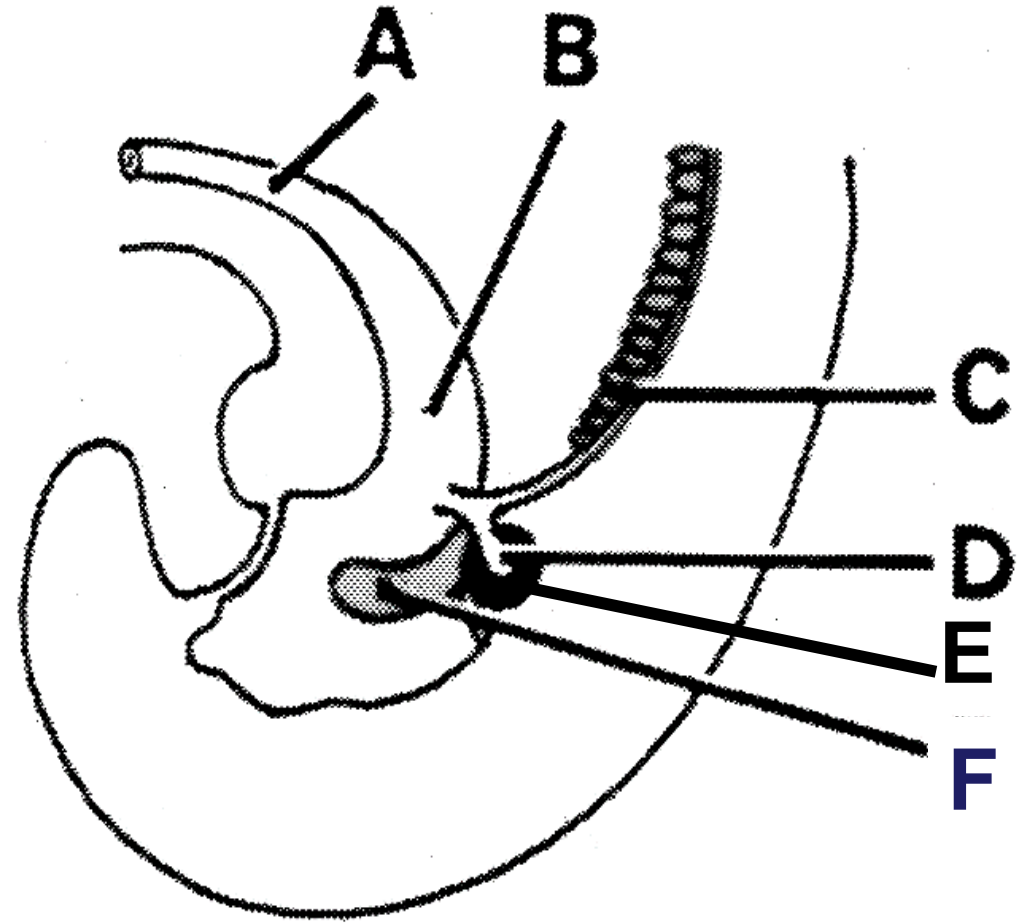
10. A 44-year-old man comes to the emergency department because of a 2-month history of excessive weight loss, left sided flank pain and hematuria. Physical examination shows pale mucous membranes and a left sided intra-abdominal mass. CT scan shows a large multilocular mass within the parenchyma of the kidney. Examination of a biopsy specimen of this mass shows a renal cell carcinoma which originated in the distal convoluted tubules. Which of the following labeled embryonic structures is the most likely source of the precursor cells of this cancer?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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- B. B
- C. C
- D. D
- E. **E**
- F. F



SOM.MK.II.BPM1.3.CPR.2.ANAT.1465 Describe the development of the kidney from the metanephric diverticulum and the metanephric mass of mesoderm.

11. A 68-year-old man comes to the physician because of a 3-month history of urinary leakage. He says he has involuntary urinary leakage, especially at night, and a constant feeling of incomplete bladder emptying. He needs to strain to start urinating and notices dribbling even after he finishes. His temperature is 36.7°C (98.1°F), pulse is 82/min, respirations are 16/min, and blood pressure is 134/82 mm Hg. Physical examination shows a soft, distended lower abdomen. A bladder ultrasonography shows a post-void residual volume of 350 mL (normal is less than 50 mL). Which of the following is the most likely diagnosis in this patient?

- A. Urachal fistula
- B. Stress incontinence
- C. Urge incontinence
- D. Overflow incontinence
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SOM.MK.II.BPM1.3.CPR.2.ANAT.1489 Define the term “stress urinary incontinence” and explain which structures may be injured during vaginal delivery to lead to this condition.

12. A 22-year-old man comes to the clinic with a 3-month history of left-sided flank pain and occasional blood in his urine. He reports that the pain worsens with prolonged standing and physical activity, and he sometimes experiences a dragging sensation in his left testicle. His temperature is 36.7°C (98.1°F), pulse is 76/min, respirations are 16/min, and blood pressure is 124/78 mm Hg. Physical examination shows mild left flank tenderness and a palpable soft mass in the left scrotum, consistent with a varicocele. Laboratory studies reveal microscopic hematuria. A Doppler ultrasound shows compression of a blood vessel between two major abdominal vessels. Which of the following vessels normally drains into the compressed vessel in this patient?

- A. Right gonadal vein
- B. Right suprarenal vein
- C. Left suprarenal vein
- D. Inferior mesenteric vein
- E. Superior mesenteric vein

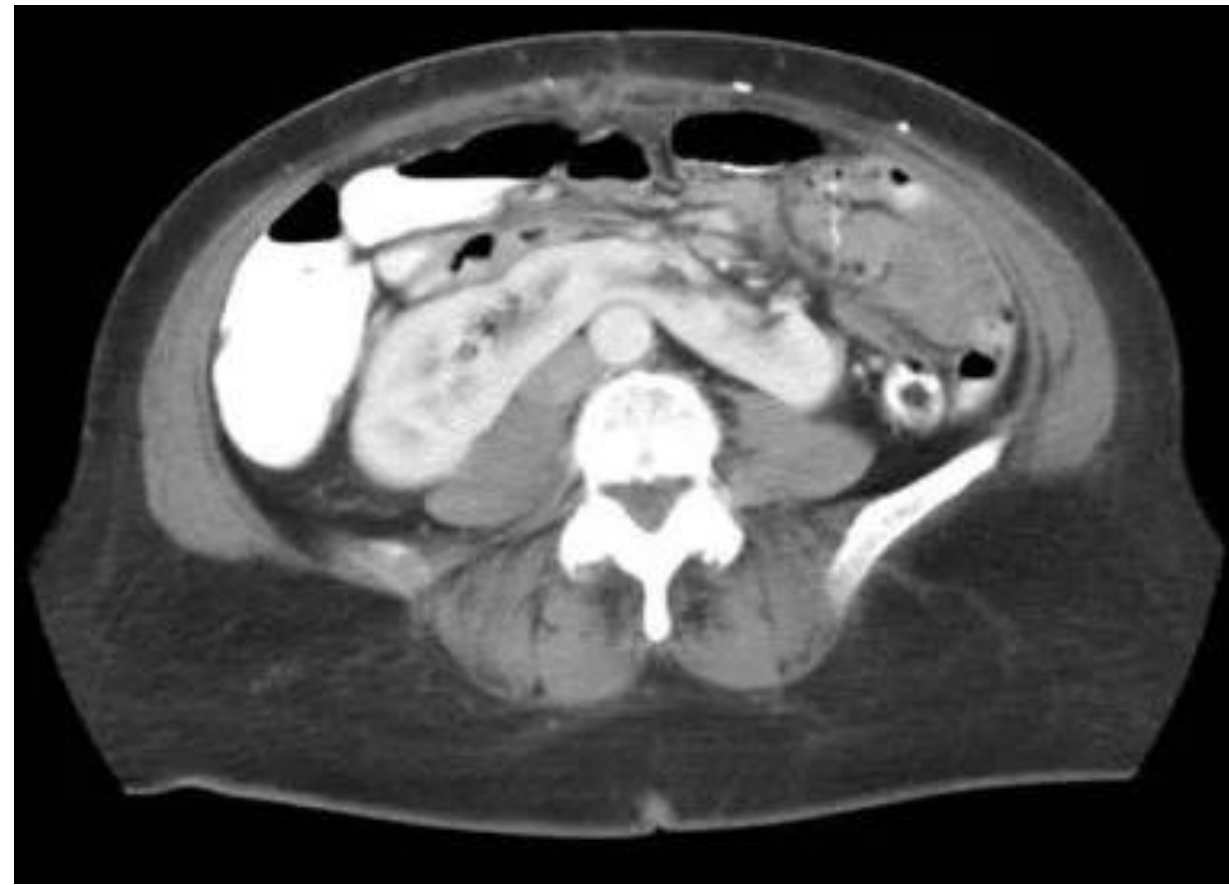
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SOM.MK.II.BPM1.3.CPR.2.ANAT.1493 Describe the anatomical location, mechanism and clinical consequences of renal vein entrapment syndrome.

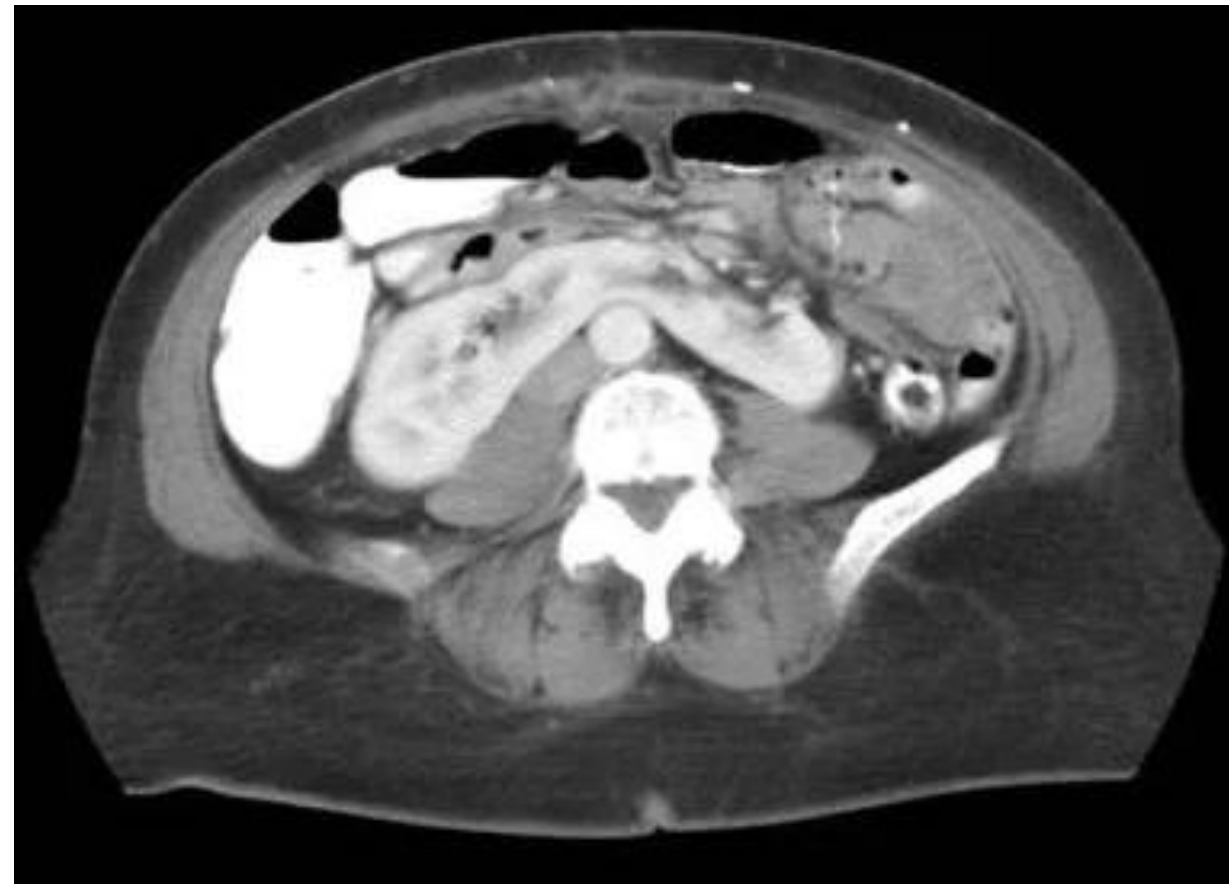
13. A 6-year-old boy is brought to the physician by his parents because of a 1-week history of fever and burning sensation while urinating. His parents mentioned that he had several urinary tract infections since birth. Temperature is 39.4°C (103°F), pulse is 92/min, respirations are 18/min, and blood pressure is 105/70 mm Hg. Physical examination shows suprapubic tenderness and costovertebral angle tenderness. Laboratory studies show a normal blood urea nitrogen and creatinine. CT scan of the abdomen is shown. Which of the following is the most likely diagnosis in this patient ?

- A. Polycystic kidney disease
- B. Horseshoe kidney
- C. Renal agenesis
- D. Bifid ureter
- E. Renal vein entrapment syndrome



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SOM.MK.II.BPM1.3.CPR.2.ANAT.1468 Describe the development of a discoid (pancake) and a horseshoe kidney.

14. A 24-year-old man is brought to the emergency room after being involved in a high-speed motor vehicle collision. He complains of intense perineal pain and an inability to urinate. His temperature is 36.9°C (98.4°F), pulse is 88/min, respirations are 18/min, and blood pressure is 122/80 mm Hg. Physical examination shows suprapubic fullness indicating a distended bladder, bruising around the perineum, and a high-riding prostate on digital rectal examination. There is blood at the external urethral meatus, and no urine is obtained despite attempted catheterization. A retrograde urethrogram demonstrates contrast extravasation from the part of the urethra that is surrounded by the external urethral sphincter. Which of the following parts of the urethra is most likely affected leading to this patient's symptoms?

- A. Spongy urethra
- B. Prostatic urethra
- C. Membranous urethra
- D. Bulbar urethra
- E. Intramural urethra

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- A. Spongy urethra
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15. A 56-year-old woman comes to the physician because of a 2-year history of recurrent urinary tract infections and persistent left lower quadrant pain. She also has occasional bouts of constipation and difficulty initiating urination. Physical examination shows mild tenderness in the left pelvic region. A CT scan of the abdomen and pelvis is shown. The left renal collecting system is most likely obstructed at which of the following anatomic locations in this patient?

- A. Urinary bladder
- B. Pelvic brim
- C. Renal calyx
- D. Ureteropelvic junction
- E. Ureterovesical junction



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SOM.MK.II.BPM1.3.CPR.2.ANAT.1476 List the sites of constriction of the ureters and explain their relationship to kidney stones

16. A 29-year-old man is brought to the emergency department after being stabbed in the left lower back during an altercation. The stab wound is located just below the 12th rib and lateral to the vertebral column. His temperature is 36.8°C (98.2°F), pulse is 90/min, respirations are 18/min, and blood pressure is 128/82 mm Hg. Physical examination shows left costovertebral angle tenderness, and a clean 3-cm stab wound in the posterior lumbar region. A non-contrast CT abdomen confirms a small hematoma external to the renal capsule. Which of the following is the last anatomical layer the knife most likely pierced before reaching the kidney?

- A. Perirenal fat
- B. Transversalis fascia
- C. Renal capsule
- D. Pararenal fat
- E. Parietal peritoneum

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